COMPLEMENTARY TURNOVER TRAINING

IN CONTINUAL REINFORCEMENT LEARNING

Problem Statement

A **big problem** in neural networks is that they suffer from **catastrophic forgetting**. This is **worsened** in **reinforcement learning environments**

What do we do?

We specifically use **CTT**, a biologically-inspired **architecture**, over **Long Short-Term Memory** neural networks, seeing **how well it counters** against **catastrophic forgetting** in **RL task sequences**

The CTT Pipeline

1 Score

Compute health/utility score for each neuron

2 Kill

Permanently remove bottom p% by score

3 Birth

Replace each killed neuron through selection

4 Gate

Newborns start silent and linearly ramped in over time

5 Homeostasis

Homeostasis controls activity stats

1 Does **Hebbian-scored** neural **turnover** during **simulated sleep cycles** enable **more** robust **memory consolidation** in an **LSTM foraging agent** in comparison to **scoring** neurons **in isolation** through CTT?

$$S(i) = \alpha \mathcal{I}(i) - \beta \mathcal{R}(i), \quad (2)$$

ere the importance and redundancy terms are

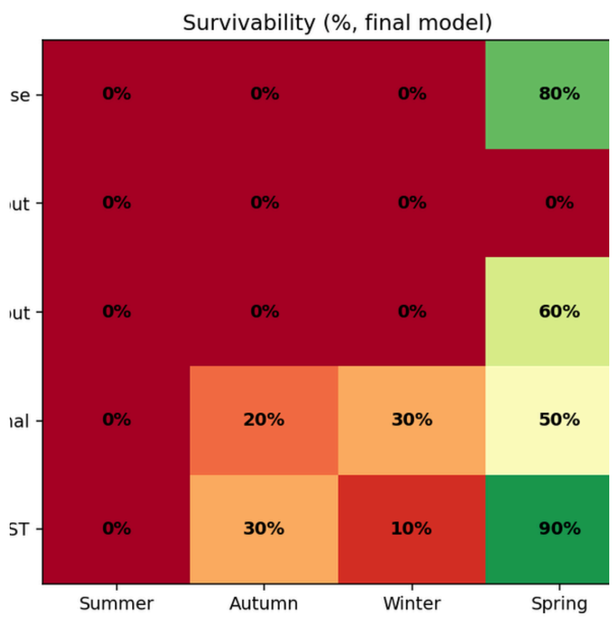
$$\mathcal{I}(i) = \overline{|a_i|} \cdot \overline{|g_i|}, \quad (3)$$

$$\mathcal{R}(i) = \max_{j \neq i} \operatorname{corr}(h_i, h_j). \quad (4)$$

Original Scorer

$$G(i) = \frac{1}{\sum_j |W_{ij}| + \epsilon} \sum_j |W_{ij}| \operatorname{corr}(x_j, a_i)$$

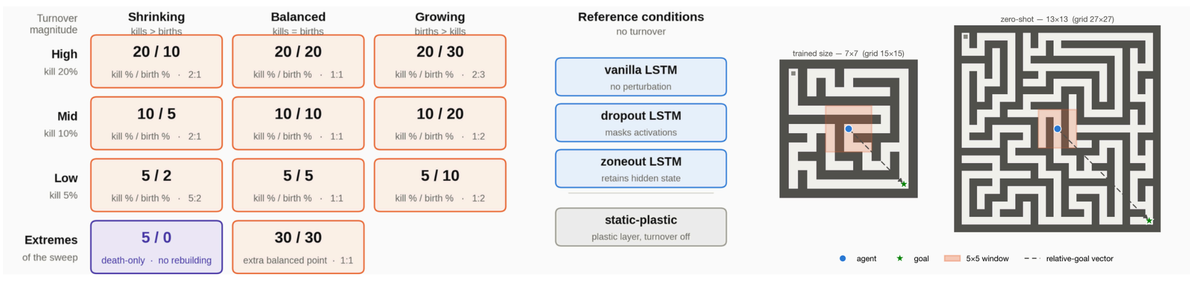
GHOST Scorer



What We Found

1. **CTT** showed proof of **remembering**
2. **Scorers** are **second order** to architecture
3. Replay + EWC + Synaptic Scaling **not sufficient** for **this** task **sequence**
4. **Future** solutions should focus on **new algorithms** for **CRL task sequences**

2 **3** Does the **kill/birth ratio** govern how well a CTT-based recurrent policy **generalises zero-shot** to mazes larger than any seen in training?



Tested ratios | Environment | Results

What We Found

1. No effect on generalisation - **Every ratio tracks baselines**
2. An initial effect was **nullified** by **more seeds (3 → 8)**
3. With our setup, CTT has built-in **inefficiency**
4. Rebirth is **vital**

NATHAN DEL FAVA - DLFNAT001@MYUCT.AC.ZA
ZIYAAD SHABOODIEN - SHBZIY001@MYUCT.AC.ZA
A/PROF GEOFF NITSCHKE - SUPERVISOR
PHD CAND. BILAL ASLAN - CO-SUPERVISOR

SCHOOL OF IT